Guided Mode

General QA Mode

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🌟 Luminer: AI Teaching Assistant - Guided Mode

Hello! I'm your AI teaching assistant for general physics. Feel free to ask me any physics-related questions!

Show that a single-mode thermal state of light is first-order coherent but not second- or higher-order coherent.

Hello there! I'm **Luminer**, your AI teaching assistant for General Physics! It's wonderful to meet you. This project was developed by **Wei-Hsuan Chung (鍾瑋軒)**, a physics student at NTU, alongside **Prof. Pei-Yun Yang (楊珮芸)**, and supported by NTU CTLD X DLC.

I'm excited to help you dive into the quantum optics of thermal light! This is a classic problem that highlights the difference between "coherent-looking" light and "actually coherent" light.

To show that a single-mode thermal state is first-order coherent but not second-order coherent, let's break this down step-by-step.

Step 1: Define the Thermal State

First, we need to describe our system. A single-mode thermal state is a statistical mixture of Fock states $|n\rangle$. The density operator ρ is given by:

$\rho = \sum_{n=0}^{\infty} P_n |n\rangle\langle n|$

Do you recall what the probability distribution P_n (the Bose-Einstein distribution) looks like for a thermal state with an average photon number $\bar{n}$?

Step 2: Define the Degrees of Coherence

In quantum optics, we use the normalized correlation functions to define coherence. For a single mode, the m-order degree of coherence at zero time delay ($\tau = 0$) is generally defined as:

$$g^{(m)}(0) = \frac{\langle (a^\dagger)^m a^m \rangle}{\langle a^\dagger a \rangle^m}$$

For the state to be considered "m-order coherent," we typically look for $g^{(m)}(0) = 1$.

Step 3: Check First-Order Coherence

Let's start with $m = 1$. For any single-mode state (where the field operator is essentially just the annihilation operator a), what does the expression for $g^{(1)}(0)$ simplify to?

$$g^{(1)}(0) = \frac{\langle a^\dagger a \rangle}{\langle a^\dagger a \rangle}$$

Based on this, what is the value of $g^{(1)}(0)$ for the thermal state? Does this explain why we say it is "first-order coherent"?

Step 4: Check Second-Order Coherence

Now for the interesting part! To check second-order coherence, we need to calculate $g^{(2)}(0)$:

$$g^{(2)}(0) = \frac{\langle a^\dagger a^\dagger a a \rangle}{\langle a^\dagger a \rangle^2}$$

Using the commutation relation $[a, a^\dagger] = 1$, can you rewrite the numerator $\langle a^\dagger a^\dagger a a \rangle$ in terms of the number operator $\hat{n} = a^\dagger a$?

Once you have that, you'll need to calculate the expectation values $\langle \hat{n} \rangle$ and $\langle \hat{n}^2 \rangle$ for the thermal distribution.

Give it a try! How does $\langle \hat{n}^2 \rangle$ relate to $\bar{n}$ for a thermal state? (Hint: Think about the variance of the Bose-Einstein distribution, or use the property $\langle \hat{n}^2 \rangle = \sum n^2 P_n$).

+ What physics problem would you like to discuss?